# Collisions Transfer Energy — Practice

Chapter 5 · Collisions Transfer Energy · Week 6

Name: _____ Date: _____

## A. WHAT CHANGED?

When objects collide, what can change? Circle ALL that can change.

**MOTION (speed/direction)** **SHAPE** **SOUND**

Give ONE example of each from the Reader.

## B. SPEED (same car)

The SAME car rolls slowly, then fast, into the SAME target.

1. Which roll moves the target farther? Why?

2. We changed only the speed. What did we keep the same? (name the car, the target, and the surface)

## C. MASS / WEIGHT (same speed)

Two cars roll at about the SAME speed into the SAME target. One car weighs more than the other.

1. Which car has more energy of motion? Why?

2. Which car moves the target farther?

## D. FAIR TEST

This plan is NOT a fair test. Fix it:

*"I changed the ramp AND the car at the same time."*

Your fix:

## E. ENERGY DOES NOT DISAPPEAR

A car hits a cup. The cup moves, and we hear a sound. Where did the car's energy of motion go?